Simplex quadrature nodal DG method for wave equations on sphere

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1. $2n-1$ and $2n$ Quadrature points in 2-simplex

1.1. Barycentric Coordinates

Consider the two-dimensional plane with coordinates $\boldsymbol{\xi} = (\xi, \eta)$. Let T be the triangle with vertices

$$\boldsymbol{\xi}_{v_1} = (0,1), \quad \boldsymbol{\xi}_{v_2} = (0,0), \quad \boldsymbol{\xi}_{v_3} = (1,0)$$

Any point within the triangle can be expressed as

$$(\xi, \eta) = \boldsymbol{\xi} = \sum_{i=0}^{3} b_i \boldsymbol{\xi}_{v_i}, \quad \sum_{i=1}^{3} b_i = 1, \quad b_i \geq 0 \tag{1}$$

where b_i for $i = 1, 2, 3$ are known as the barycentric coordinates.

Numerical integration is often encounter in computations. To conduct numerical integration over T, numerical quadrature points and weights are introduced in []. Below we quote the barycentric coordinates of the quadrature points and the associated weights for conducting integration of polynomials of degree at most degree $2n-1$ and $2n$, in Table 1 and Table 2.

n	Sym	b_1	b_2	w_i^s	w_i^e
1	S_6	0.2113248654051871	0.0000000000000000	0.16666666666666667	0.5000000000000000
2	S_6	0.1127016653792583	0.0000000000000000	0.04166666666666666	0.2777777777777777
2	S_3	0.5000000000000000	0.0000000000000000	0.09999999999999999	0.4444444444444444
2	S_1	0.3333333333333333	0.3333333333333333	0.45000000000000000	–
3	S_6	0.06943184420297367	0.0000000000000000	0.01509901487256561	0.1739274225687269
3	S_6	0.3300094782075718	0.0000000000000000	0.04045654068298990	0.3260725774312731
3	S_6	0.5841571139756568	0.1870738791912763	0.11111111111111111	–
4	S_6	0.04691007703066797	0.0000000000000000	0.006601315081001592	0.1184634425280944
4	S_6	0.2307653449471584	0.0000000000000000	0.02053045968042892	0.2393143352496833
4	S_3	0.5000000000000000	0.0000000000000000	0.01853708483394990	0.2844444444444446
4	S_3	0.1394337314154536	0.1394337314154536	0.10542932962084440	–
4	S_3	0.4384239524408185	0.4384239524408185	0.12473673228977350	–
4	S_1	0.3333333333333333	0.3333333333333333	0.09109991119771331	–

Table 1: Barycentric coordinates of quadrature integration points for integrating polynomials of degree at most $2n-1$ in a triangle. w_i^s are surface integration associated quadrature weights. w_i^e are line integration quadrature weights associated with edge points.

n	Sym	b_1	b_2	w_i^s
1	S_3	0.1666666666666666	0.1666666666666666	0.3333333333333333
2	S_3	0.09157621350977067	0.09157621350977067	0.1099517436553218
2	S_3	0.4459484909159648	0.4459484909159648	0.2233815896780115
3	S_3	0.219429982549783	0.219429982549783	0.1713331241529809
3	S_3	0.480137964112215	0.480137964112215	0.08073108959303095
3	S_6	0.1416190159239682	0.0193717243612408	0.04063455979366068
4	S_6	0.7284923929554044	0.2631128296346379	0.02723031417443505
4	S_3	0.4592925882927232	0.4592925882927232	0.09509163426728455
4	S_3	0.1705693077517602	0.1705693077517602	0.1032173705347182
4	S_3	0.05054722831703096	0.05054722831703096	0.03245849762319804
4	S_1	0.3333333333333333	0.3333333333333333	0.1443156076777874

Table 2: Barycentric coordinates of quadrature integration points for integrating polynomials of degree at most $2n$ in a triangle. w_i^s are surface integration associated quadrature weights.

Based on these points we evaluate surface integration over a triangle, with $|T|$ being the area of the triangle, by a discrete summation as follows,

$$\int_T f(\boldsymbol{\xi})\, d\boldsymbol{\xi} = |T| \sum_{i=1}^{N} f(\boldsymbol{\xi}_i) w_i^s, \tag{2}$$

provided that f is a polynomial of degree at most $2n-1$ and $2n$. For evaluating line integrals on an edge of a triangle, we adopt the following quadrature formula:

$$\int_{\partial T^\gamma} f(\boldsymbol{\xi})\, d\boldsymbol{\xi} = |\partial T^\gamma| \sum_{i=1}^{n+1} f(\boldsymbol{\xi}_i^\gamma) w_i^e \tag{3}$$

where $|\partial T^\gamma|$ denotes the length of the edge γ, $\boldsymbol{\xi}_i^\gamma$ are the quadrature points on the edge, and w_i^e are the associated quadrature weights.

2. Orthogonal Polynomials on Simplex

We denote the triangle with vertices $(-1,1)$, $(-1,-1)$, and $(1,-1)$ by

$$\mathsf{T}^2 = \{(r,s) \,|\, r,s \geq -1,\ r+s \leq 1\}.$$

Let $n \in \{1,2,3,4\}$. On T^2, let us introduce the orthogonal polynomials $\phi_{i,j}$ defined by

$$\phi_m(r,s) = \Phi_{ij}(r,s) = \sqrt{2} P_i(a) P_j^{(2i+1,j)}(b)(1-b)^j, \quad 0 \leq i,j \leq n, \quad i+j \leq n,$$
$$m = j + (n+1)i + 1 - \frac{i}{2}(i-1), \quad a = 2 \cdot \frac{1+r}{1-s} - 1, \quad b = s$$

where P_i is the Legendre polynomial of degree i and $P_j^{(2i+1,j)}$ is the Jabobi polynomials of degree j. Applying these basis functions we span the function space as

$$\mathcal{V} = \text{span}\{\Phi_{ij}(r,s)\,|\,0 \le i,j \le n, \quad i+j \le n\}$$

Thus, for $n = 1$, we have the basis functions

$$\begin{aligned}
\frac{\phi_1(r,s)}{\sqrt{2}} &= \frac{\Phi_{00}}{\sqrt{2}} = P_0\left(\frac{2(1+r)}{1-s}\right) P_0^{(1,0)}(s)(1-s)^0 = 1 \\
\frac{\phi_2(r,s)}{\sqrt{2}} &= \frac{\Phi_{01}}{\sqrt{2}} = P_0\left(\frac{2(1+r)}{1-s}\right) P_1^{(1,1)}(s)(1-s)^1 = 2 + 4\frac{s-1}{2}(1-s)^1 = 2 - 2(1-s)^2 \\
\frac{\phi_3(r,s)}{\sqrt{2}} &= \frac{\Phi_{10}}{\sqrt{2}} = P_1\left(\frac{2(1+r)}{1-s}\right) P_0^{(3,0)}(s)(1-s)^0 = 1 + \left(\frac{2(1+r)}{1-s} - 1\right) = 2\frac{1+r}{1-s}
\end{aligned}$$

For $n = 2$, we have six functions,

$$\begin{aligned}
\frac{\phi_1(r,s)}{\sqrt{2}} &= \frac{\Phi_{00}}{\sqrt{2}} = P_0\left(\frac{2(1+r)}{1-s}\right) P_0^{(1,0)}(s)(1-s)^0 = 1 \\
\frac{\phi_2(r,s)}{\sqrt{2}} &= \frac{\Phi_{01}}{\sqrt{2}} = P_0\left(\frac{2(1+r)}{1-s}\right) P_1^{(1,1)}(s)(1-s)^1 = 2 - 2(1-s)^2 \\
\frac{\phi_3(r,s)}{\sqrt{2}} &= \frac{\Phi_{02}}{\sqrt{2}} = P_0\left(\frac{2(1+r)}{1-s}\right) P_2^{(1,0)}(s)(1-s)^2 = \left(3 + 6\left(\frac{2(1+r)}{1-s} - 1\right) + \frac{5}{2}\left(\frac{2(1+r)}{1-s} - 1\right)^2\right)(1-s)^2 \\
\frac{\phi_4(r,s)}{\sqrt{2}} &= \frac{\Phi_{10}}{\sqrt{2}} = P_1\left(\frac{2(1+r)}{1-s}\right) P_0^{(3,0)}(s)(1-s)^0 = 2\frac{1+r}{1-s} \\
\frac{\phi_5(r,s)}{\sqrt{2}} &= \frac{\Phi_{11}}{\sqrt{2}} = P_1\left(\frac{2(1+r)}{1-s}\right) P_1^{(3,1)}(s)(1-s)^1 = 2\frac{1+r}{1-s}\left(4 + 3\left(\frac{2(1+r)}{1-s} - 1\right)\right)(1-s)^1 \\
\frac{\phi_6(r,s)}{\sqrt{2}} &= \frac{\Phi_{20}}{\sqrt{2}} = P_2\left(\frac{2(1+r)}{1-s}\right) P_0^{(5,0)}(s)(1-s)^0 = 1 + 3\left(\frac{2(1+r)}{1-s} - 1\right) + \frac{3}{2}\left(\frac{2(1+r)}{1-s} - 1\right)^2
\end{aligned}$$

3. Approximation, Integration, and Differentiation

3.1. Modal and nodal approximation

Based on these grid points, we may seek numerical approximations to a function u defined on T.

Consider a function space $\mathcal{V}$ spanned by a set of linearly independent basis functions $\phi_k(\xi,\eta)$ for $i = 1, 2, \cdots, n$, i.e.,

$$\mathcal{V} = \text{span}\{\phi_i(\xi,\eta)|i = 1, 2, ..., n\}.$$

Thus, a function $u(\xi,\eta) \in \mathcal{V}$ can be approximated as follows,

$$u(\xi,\eta) = \sum_{i=1}^{N} a_i \phi_i(\xi,\eta), \tag{4}$$

where a_i are coefficients. Applying the quadrature points (ξ_m, η_m) for $m = 1, 2, \cdots, M$ and denoting $u(\xi_m, \eta_m) = u_m$ for simplicity, we have

$$\boldsymbol{u} = \boldsymbol{V}\boldsymbol{a}, \tag{5}$$

where

$$\boldsymbol{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_M \end{bmatrix}, \quad \boldsymbol{V} = \begin{bmatrix} \phi_1(\xi_1,\eta_1) & \phi_2(\xi_1,\eta_1) & \cdots & \phi_N(\xi_1,\eta_1) \\ \phi_1(\xi_2,\eta_2) & \phi_2(\xi_2,\eta_2) & \cdots & \phi_N(\xi_2,\eta_2) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_1(\xi_M,\eta_M) & \phi_2(\xi_M,\eta_M) & \cdots & \phi_N(\xi_M,\eta_M) \end{bmatrix}, \quad \boldsymbol{a} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_N \end{bmatrix}. \tag{6}$$

Let

$$\boldsymbol{W} = \mathrm{diag}(w_1^s, w_2^s, \cdots, w_M^s) \tag{7}$$

Define the mass matrix over the triangle T as

$$\boldsymbol{M} = |T|\boldsymbol{V}^T\boldsymbol{W}\boldsymbol{V}. \tag{8}$$

Since $\boldsymbol{W}$ is diagonal, we have from Eq. (8)

$$\boldsymbol{M}^T = |T|(\boldsymbol{V}^T\boldsymbol{W}\boldsymbol{V})^T = |T|\boldsymbol{V}^T\boldsymbol{W}\boldsymbol{V} = \boldsymbol{M} \tag{9}$$

Furthermore, for $\boldsymbol{a}$ being a non-zero vector and $\boldsymbol{W}$ being positive definite, we have

$$\boldsymbol{a}^T\boldsymbol{M}\boldsymbol{a} = |T|(\boldsymbol{V}\boldsymbol{a})^T\boldsymbol{W}(\boldsymbol{V}\boldsymbol{a}) > 0. \tag{10}$$

indicating that $\boldsymbol{M}$ is not only symmetric but also positive definite. Thus, its inverse exists, denoted by $\boldsymbol{M}^{-1}$.

Then, multiplying $|T|\boldsymbol{V}^T\boldsymbol{W}$ to Eq. (5), we have the vector $\boldsymbol{a}$ as follows,

$$|T|\boldsymbol{V}^T\boldsymbol{W}\boldsymbol{u} = |T|\boldsymbol{V}^T\boldsymbol{W}\boldsymbol{V}\boldsymbol{a} = \boldsymbol{M}\boldsymbol{a} \implies \boldsymbol{a} = |T|\boldsymbol{M}^{-1}\boldsymbol{V}^T\boldsymbol{W}\boldsymbol{u} \tag{11}$$

3.2. Numerical differentiation

To perform numerical differentiation on the vector $\boldsymbol{u}$, we need to construct the differentiation matrices for differentiating functions in the ξ and η directions. Differentiating Eq. (4) with respect to ξ, we have

$$\partial_\xi u(\xi,\eta) = \sum_{i=1}^{N} a_i \partial_\xi \phi(\xi,\eta) \tag{12}$$

Hence, the grid vector version of the above expression is

$$\boldsymbol{u}_\xi = \boldsymbol{V}_\xi \boldsymbol{a}, \quad \boldsymbol{u}_\xi = \begin{bmatrix} \partial_\xi u_1 \\ \partial_\xi u_2 \\ \vdots \\ \partial_\xi u_M \end{bmatrix} \quad \boldsymbol{V}_\xi = \begin{bmatrix} \partial_\xi\phi_1(\xi_1,\eta_1) & \partial_\xi\phi_2(\xi_1,\eta_1) & \cdots & \partial_\xi\phi_N(\xi_1,\eta_1) \\ \partial_\xi\phi_1(\xi_2,\eta_2) & \partial_\xi\phi_2(\xi_2,\eta_2) & \cdots & \partial_\xi\phi_N(\xi_2,\eta_2) \\ \vdots & \vdots & \ddots & \vdots \\ \partial_\xi\phi_1(\xi_M,\eta_M) & \partial_\xi\phi_2(\xi_M,\eta_M) & \cdots & \partial_\xi\phi_N(\xi_M,\eta_M) \end{bmatrix} \tag{13}$$

Applying the relationship Eq. (11) we have

$$\boldsymbol{u}_\xi = \boldsymbol{V}_\xi \boldsymbol{a} = |T| \boldsymbol{V}_\xi \boldsymbol{M}^{-1} \boldsymbol{V}^T \boldsymbol{W} \boldsymbol{u} = \boldsymbol{D}_\xi \boldsymbol{u} \tag{14}$$

where $\boldsymbol{D}_\xi$ is the differentiation matrix in the ξ direction. Likewise, we construct the differentiation matrix in the η direction and perform the associated numerical differentiation as follows,

$$\boldsymbol{D}_\eta = |T| \boldsymbol{V}_\eta \boldsymbol{M}^{-1} \boldsymbol{V}^T \boldsymbol{W}, \quad \boldsymbol{u}_\eta = \boldsymbol{D}_\eta \boldsymbol{u}. \tag{15}$$

3.3. Numerical Integration

Here we briefly summarize the extension of the quadrature rule.

Let $q(\boldsymbol{\xi})$ be a scalar function and $\boldsymbol{V}(\boldsymbol{\xi}) = (u(\boldsymbol{\xi}), v(\boldsymbol{\xi}))$ be a vector-valued function. From the divergence theorem, we have

$$\begin{aligned} \oint_{\partial \mathcal{T}} \boldsymbol{n} \cdot (q\boldsymbol{F})\, d\boldsymbol{\xi} &= \int_{\mathcal{T}} \nabla \cdot (q\boldsymbol{F})\, d\boldsymbol{\xi} \\ &= \int_{\mathcal{T}} q \nabla \cdot \boldsymbol{F}\, d\boldsymbol{\xi} + \int_{\mathcal{T}} \boldsymbol{F} \cdot \nabla q\, d\boldsymbol{\xi} \end{aligned} \tag{16}$$

If both q and $\boldsymbol{F}$ are polynomials of degree at most n then we can apply the simplex quadrature rule to establish discrete version of the equation as follows,

$$(\boldsymbol{q}^e)^T \boldsymbol{W}^e \boldsymbol{f}_n^e = |T| \boldsymbol{1}^T \boldsymbol{W} \left(\boldsymbol{D}_\xi (\boldsymbol{q} \odot \boldsymbol{u}) + \boldsymbol{D}_\eta (\boldsymbol{q} \odot \boldsymbol{v}) \right) \tag{17}$$

$$= |T| \boldsymbol{q}^T \boldsymbol{W} (\boldsymbol{D}_\xi \boldsymbol{u} + \boldsymbol{D}_\eta \boldsymbol{v}) + |T| (\boldsymbol{u}^T \boldsymbol{W} \boldsymbol{D}_\xi \boldsymbol{q} + \boldsymbol{v}^T \boldsymbol{W} \boldsymbol{D}_\eta \boldsymbol{q}) \tag{18}$$

where $\boldsymbol{q}^e$ and $\boldsymbol{f}_n^e$ are grid vector representations of the functions q and $\boldsymbol{n} \cdot F$ at the three edges, and $\boldsymbol{W}^e$ is the quadrature weight matrix with diagonal entries being w_i^e for $i = 1, 2, ..., 2n$,

$$\boldsymbol{W}^e = \begin{bmatrix} |\partial T^1| & 0 & 0 \\ 0 & |\partial T^2| & 0 \\ 0 & 0 & |\partial T^3| \end{bmatrix} \otimes \mathrm{diag}(w_1^e, w_2^e, \cdots, w_{n+1}^e)$$

3.4. Well-posed analysis for model wave equations

Let us consider applying discontinuous Galerkin method to solve the model wave equation:

$$\partial_t q(\boldsymbol{\xi}, t) + \nabla \cdot (\boldsymbol{V} q(\boldsymbol{\xi}, t)) = 0, \quad \boldsymbol{\xi} \in \mathcal{T} \tag{19}$$

$$q(\boldsymbol{\xi}, 0) = q_0(\boldsymbol{\xi}), \quad \boldsymbol{\xi} \in \mathcal{T} \tag{20}$$

$$q(\boldsymbol{\xi}, t) = g(\boldsymbol{\xi}, t), \quad \text{if } \boldsymbol{n} \cdot \boldsymbol{V} < 0 \text{ for } \boldsymbol{\xi} \in \gamma \subset \partial \mathcal{T}, \quad t \geq 0 \tag{21}$$

We now conduct an energy estimate of the problem. The governing equation can be expressed in the following split form:

$$\partial_t q = -\frac{1}{2} \nabla \cdot (\boldsymbol{V} q) - \frac{1}{2} \boldsymbol{V} \cdot \nabla q - \frac{1}{2} (\nabla \cdot \boldsymbol{V}) q \tag{22}$$

Multiplying $2q$ to the equation, integrating over the domain $\mathcal{T}$, and invoking the identity $\nabla \cdot (\boldsymbol{V} q^2) = q\nabla \cdot (\boldsymbol{V} q) + q\boldsymbol{V} \cdot \nabla q$ as well the divergence theorem, we have

$$\begin{aligned}\frac{d}{dt}\int_{\mathcal{T}} q^2(\boldsymbol{\xi}, t)\, d\boldsymbol{\xi} &= -\int_{\mathcal{T}} q\nabla \cdot (\boldsymbol{V} q) + q\boldsymbol{V} \cdot \nabla q + (\nabla \cdot \boldsymbol{V}) q^2\, d\boldsymbol{\xi} \\ &= -\int \nabla \cdot (\boldsymbol{V} q^2) + (\nabla \cdot \boldsymbol{V}) q^2\, d\boldsymbol{\xi} \\ &= -\oint_{\partial\mathcal{T}} (n \cdot \boldsymbol{V}) q^2 d\boldsymbol{\xi} - \int_{\mathcal{T}} (\nabla \cdot \boldsymbol{V}) q^2\, d\boldsymbol{\xi} \end{aligned} \tag{23}$$

Let $\alpha = \max_{\boldsymbol{\xi} \in \mathcal{T}} |\nabla \cdot \boldsymbol{V}|$.

$$\frac{d}{dt}\int_{\mathcal{T}} q^2(\boldsymbol{\xi}, t)\, d\boldsymbol{\xi} - \alpha \int_{\mathcal{T}} q^2\, d\boldsymbol{\xi} \leq \int_{\gamma} (-n \cdot \boldsymbol{V}) g^2\, d\boldsymbol{\xi} \tag{24}$$

Multiplying $e^{-\alpha t}$ to the equation and integrating with respect to t, we have

$$e^{-\alpha t}\int_{\mathcal{T}} q^2(\boldsymbol{\xi}, t)\, d\boldsymbol{\xi} - \int_{\mathcal{T}} q^2(\boldsymbol{\xi}, 0)\, d\boldsymbol{\xi} \leq \int_0^t \int_{\gamma} (-n \cdot \boldsymbol{V}) g^2\, d\boldsymbol{\xi}\, dt \tag{25}$$

leading to

$$\int_{\mathcal{T}} q^2(\boldsymbol{\xi}, t)\, d\boldsymbol{\xi} \leq e^{\alpha t}\left(\int_{\mathcal{T}} q_0^2(\boldsymbol{\xi})\, d\boldsymbol{\xi} + \int_0^t \int_{\gamma} (-n \cdot \boldsymbol{V}) g^2(t')\, d\boldsymbol{\xi}\, dt'\right) \tag{26}$$

3.5. DG scheme for advection equation

We now proceed constructing a DG scheme for solving advection equation.

We consider the following discretization:

$$\begin{aligned}\partial_t \boldsymbol{q} = &-\frac{1}{2}\left(\boldsymbol{D}_\xi(\boldsymbol{u} \odot \boldsymbol{q}) + \boldsymbol{D}_\eta(\boldsymbol{v} \odot \boldsymbol{q})\right) - \frac{1}{2}(\boldsymbol{u} \odot \boldsymbol{D}_\xi \boldsymbol{q} + \boldsymbol{v} \odot \boldsymbol{D}_\eta \boldsymbol{q}) - \frac{1}{2}(\boldsymbol{D}_\xi \boldsymbol{u} + \boldsymbol{D}_\eta \boldsymbol{v}) \odot \boldsymbol{q} \\ &+ (|T|)^{-1}\boldsymbol{W}^{-1}\boldsymbol{E}^T\boldsymbol{W}^e\boldsymbol{p}\end{aligned} \tag{27}$$

where

$$\boldsymbol{E} = \left[\boldsymbol{I}_{3(n+1)\times 3(n+1)} | \boldsymbol{0}_{3(n+1)\times(M-3(n+1))}\right] \tag{28}$$

and $\boldsymbol{p}$ is the penalized boundary condition vector, and it is form is to be determined later.

Multiplying $2|T|\boldsymbol{q}^T\boldsymbol{W}$ from the left to the scheme and invoking the discrete divergence theorem, we have

$$\begin{aligned}\frac{d}{dt}(|T|\boldsymbol{q}^T\boldsymbol{W}\boldsymbol{q}) = &-|T|\boldsymbol{q}^T\boldsymbol{W}\left(\boldsymbol{D}_\xi(\boldsymbol{u} \odot \boldsymbol{q}) + \boldsymbol{D}_\eta(\boldsymbol{v} \odot \boldsymbol{q})\right) - |T|\boldsymbol{q}^T\boldsymbol{W}(\boldsymbol{u} \odot \boldsymbol{D}_\xi \boldsymbol{q} + \boldsymbol{v} \odot \boldsymbol{D}_\eta \boldsymbol{q}) \\ &- |T|\boldsymbol{q}^T\boldsymbol{W}(\boldsymbol{D}_\xi \boldsymbol{u} + \boldsymbol{D}_\eta \boldsymbol{v}) \odot \boldsymbol{q} + 2|T|(|T|)^{-1}\boldsymbol{q}^T\boldsymbol{W}\boldsymbol{W}^{-1}\boldsymbol{E}^T\boldsymbol{W}^e\boldsymbol{p} \\ = &-\oint_{\partial\mathcal{T}} (\boldsymbol{n} \cdot \boldsymbol{V}) q^2\, d\boldsymbol{x} + |T|(\boldsymbol{u} \odot \boldsymbol{q})^T\boldsymbol{W}\boldsymbol{D}_\xi \boldsymbol{q} + |T|(\boldsymbol{v} \odot \boldsymbol{q})^T\boldsymbol{W}\boldsymbol{D}_\eta \boldsymbol{q} \\ &- |T|\boldsymbol{q}^T\boldsymbol{W}(\boldsymbol{u} \odot \boldsymbol{D}_\xi \boldsymbol{q} + \boldsymbol{v} \odot \boldsymbol{D}_\eta \boldsymbol{q}) - |T|\boldsymbol{q}^T\boldsymbol{W}(\boldsymbol{D}_\xi \boldsymbol{u} + \boldsymbol{D}_\eta \boldsymbol{v}) \odot \boldsymbol{q} + 2\boldsymbol{q}^T\boldsymbol{E}^T\boldsymbol{W}^e\boldsymbol{p}\end{aligned} \tag{29}$$

Note that

$$(\boldsymbol{u} \odot \boldsymbol{q})^T \boldsymbol{W} \boldsymbol{D}_\xi \boldsymbol{q} = \sum_{i=0}^{N} u_i q_i (\boldsymbol{D}_\xi \boldsymbol{q})_i w_i^s = \boldsymbol{q}^T \boldsymbol{W} (\boldsymbol{u} \odot \boldsymbol{D}_\xi \boldsymbol{q}), \tag{30}$$

$$(\boldsymbol{v} \odot \boldsymbol{q})^T \boldsymbol{W} \boldsymbol{D}_\eta \boldsymbol{q} = \sum_{i=0}^{N} v_i q_i (\boldsymbol{D}_\eta \boldsymbol{q})_i w_i^s = \boldsymbol{q}^T \boldsymbol{W} (\boldsymbol{c} \odot \boldsymbol{D}_\xi \boldsymbol{q}), \tag{31}$$

$$2\boldsymbol{q}^T \boldsymbol{E}^T \boldsymbol{W}^e \boldsymbol{p} = 2(\boldsymbol{q}^e)^T \boldsymbol{W} \boldsymbol{p} \tag{32}$$

Hence,

$$\frac{d}{dt}(|T| \boldsymbol{q}^T \boldsymbol{W} \boldsymbol{q}) = -\oint_{\partial \mathcal{T}} (\boldsymbol{n} \cdot \boldsymbol{V}) q^2 \, d\boldsymbol{x} - |T| \boldsymbol{q}^T \boldsymbol{W} ((\boldsymbol{D}_\xi \boldsymbol{u} + \boldsymbol{D}_\eta \boldsymbol{v}) \odot \boldsymbol{q}) + 2(\boldsymbol{q}^e)^T \boldsymbol{W}^e \boldsymbol{p} \tag{33}$$

Consider the boundary term as

$$2(\boldsymbol{q}^e)^T \boldsymbol{W}^e \boldsymbol{p} = \oint_{\partial \mathcal{T}} 2q \left(\boldsymbol{n} \cdot (\boldsymbol{f} - \boldsymbol{f}^*) \right) d\boldsymbol{\xi} \tag{34}$$

where

$$\boldsymbol{f}(q^-) = \boldsymbol{V} q^-, \tag{35}$$

$$\boldsymbol{f}^*(q^-, q^+) = \frac{\boldsymbol{V} q^- + \boldsymbol{V} q^+}{2} + (1 - \tau) \frac{|\boldsymbol{n}^- \cdot \boldsymbol{V}|}{2} (\boldsymbol{n}^- q^- + \boldsymbol{n}^+ q^+) \tag{36}$$

where $0 \leq \tau \leq 1$.

3.6. Numerical boundary and interface conditions

3.6.1. Bourdary condition by upwind flux ($\tau = 0$, $q^- = q$ and $q^+ = g$)

To enforce boundary condition, we have, for $q^- = q$, $q^+ = g$, $\boldsymbol{n}^- = \boldsymbol{n}$, $\boldsymbol{n}^+ = -\boldsymbol{n}$,

$$\boldsymbol{f}(q) = \boldsymbol{V} q,$$

$$\boldsymbol{f}^*(q, g) = \frac{\boldsymbol{V} q + \boldsymbol{V} g}{2} + \frac{|\boldsymbol{n} \cdot \boldsymbol{V}|}{2} (\boldsymbol{n} q - \boldsymbol{n} g)$$

We now discuss the impact of the numerical flux f^*.

Case 1: out flow: $\boldsymbol{n} \cdot \boldsymbol{V} \geq 0$.

we have

$$q \boldsymbol{n} \cdot \boldsymbol{f}^* = \frac{\boldsymbol{n} \cdot \boldsymbol{V} q^2 + \boldsymbol{n} \cdot \boldsymbol{V} q g}{2} + \frac{\boldsymbol{n} \cdot \boldsymbol{V}}{2} (q^2 - q g) = \boldsymbol{n} \cdot \boldsymbol{V} q^2, \tag{37}$$

$$2q \boldsymbol{n} \cdot (\boldsymbol{f} - \boldsymbol{f}^*) = 2q \boldsymbol{n} \cdot \boldsymbol{f} - 2q \boldsymbol{n} \cdot \boldsymbol{f}^* = 2(\boldsymbol{n} \cdot \boldsymbol{V}) q^2 - 2(\boldsymbol{n} \cdot \boldsymbol{V}) q^2 = 0 \tag{38}$$

Case 2: inflow: $\boldsymbol{n} \cdot \boldsymbol{V} < 0$, $\boldsymbol{n}^- = \boldsymbol{n}$, $\boldsymbol{n}^+ = -\boldsymbol{n}$.

$$q \boldsymbol{n} \cdot \boldsymbol{f}^* = \frac{\boldsymbol{n} \cdot \boldsymbol{V} q^2 + \boldsymbol{n} \cdot \boldsymbol{V} q g}{2} - \frac{\boldsymbol{n} \cdot \boldsymbol{V}}{2} (q^2 - q g) = \boldsymbol{n} \cdot \boldsymbol{V} q g$$

and

$$2q\boldsymbol{n}\cdot(\boldsymbol{f}-\boldsymbol{f}^*) = 2q\boldsymbol{n}\cdot\boldsymbol{f} - 2q\boldsymbol{n}\cdot\boldsymbol{f}^* = 2(\boldsymbol{n}\cdot\boldsymbol{V})(q^2-qg)$$

Thus, the boundary term becomes

$$\oint_{\partial\mathcal{T}} \overbrace{(-\boldsymbol{n}\cdot\boldsymbol{V})q^2 + 2q\boldsymbol{n}\cdot(\boldsymbol{f}-\boldsymbol{f}^*)}^{\rho}\, d\boldsymbol{x} \leq \int_\gamma (-\boldsymbol{n}\cdot\boldsymbol{V})g^2\, d\boldsymbol{\xi} \tag{39}$$

since

$$\boldsymbol{\rho} = \begin{cases} (-\boldsymbol{n}\cdot\boldsymbol{V})q^2 \leq 0 & \text{for out flowing } (\boldsymbol{n}\cdot\boldsymbol{V}>0) \\ (\boldsymbol{n}\cdot\boldsymbol{V})(q-g)^2 - \boldsymbol{n}\cdot\boldsymbol{V}g^2 \leq (-\boldsymbol{n}\cdot\boldsymbol{V})g^2 & \text{for in flowing } (\boldsymbol{n}\cdot\boldsymbol{V}<0) \end{cases} \tag{40}$$

If $\max_{\boldsymbol{\xi}\in\mathcal{T}} ||-\boldsymbol{D}_\xi\boldsymbol{u} - \boldsymbol{D}_\eta\boldsymbol{v}|| = \alpha$ then we have the energy rate as

$$|T|\frac{d}{dt}(|T|\boldsymbol{q}^T\boldsymbol{W}\boldsymbol{q}) + \alpha|T|(\boldsymbol{q}^T\boldsymbol{W}\boldsymbol{q}) \leq -\oint_\gamma (\boldsymbol{n}\cdot\boldsymbol{V})g^2\, d\boldsymbol{\xi} \tag{41}$$

Multiplying the above expression with respect to time and applying initial condition, we have

$$(|T|\boldsymbol{q}^T\boldsymbol{W}\boldsymbol{q})(t) \leq e^{\alpha t}\left(|T|\boldsymbol{q}_0^T\boldsymbol{W}\boldsymbol{q}_0 + \int_0^t\int_{\partial T}(-\boldsymbol{n}\cdot\boldsymbol{V})g^2(\boldsymbol{\xi},t')\, d\boldsymbol{\xi}\, dt'\right) \tag{42}$$

We are left with the exact form of $\boldsymbol{p}$. Let

$$\boldsymbol{q} = [\overbrace{q_1, q_2, \cdots, q_{3(n+1)}}^{\text{boundary points}}, \overbrace{q_{3(n+1)+1}, \cdots, q_N}^{\text{interior points}}]^T, \quad \boldsymbol{p} = [\overbrace{p_1, p_2, \cdots p_{3(n+1)}}^{\text{boundary points}}]^T. \tag{43}$$

We require that

$$\begin{aligned} 2(\boldsymbol{q}^e)^T\boldsymbol{W}^e\boldsymbol{p} &= \sum_{i=1}^{n+1} 2|\partial T^1|q_i w_i^e p_i + \sum_{i=1}^{n+1} 2|\partial T^2|q_{n+1+i}w_i^e p_{n+1+i} + \sum_{i=1}^{n+1} 2|\partial T^2|q_{2(n+1)+i}w_i^e p_{2(n+1)+i} \\ &= \oint_{\partial T} 2q\boldsymbol{n}\cdot(\boldsymbol{f}-\boldsymbol{f}^*)\, d\boldsymbol{x} \\ &= \sum_{i=1}^{n+1} 2|\partial T^1|q_i(\boldsymbol{n}\cdot(\boldsymbol{f}-\boldsymbol{f}^*))\Big|_i w_i^e + \sum_{i=1}^{n+1} 2|\partial T^2|q_{n+1+i}(\boldsymbol{n}\cdot(\boldsymbol{f}-\boldsymbol{f}^*))\Big|_{n+1+i} w_i^e \\ &\quad + \sum_{i=1}^{n+1} 2|\partial T^3|q_{2(n+1)+i}(\boldsymbol{n}\cdot(\boldsymbol{f}-\boldsymbol{f}^*))\Big|_{2(n+1)+i} w_i^e \end{aligned} \tag{44}$$

Thus,

$$p_i = (\boldsymbol{n}\cdot(\boldsymbol{f}-\boldsymbol{f}^*))\Big|_i \tag{45}$$

where $f_i = f(\boldsymbol{\xi}_i)$ and $f_i^*(\boldsymbol{\xi}_i)$.

3.6.2. interface condition by central flux ($\tau = 1$, $q^- = q$ and $q^+ = g$)

The energy rate equations are

$$\begin{aligned}\frac{d}{dt}(|T|(\boldsymbol{q}^{(1)})^T\boldsymbol{W}\boldsymbol{q}^{(1)}) = &-\oint_{\partial\mathcal{T}}(\boldsymbol{n}^{(1)}\cdot\boldsymbol{V})(q^{(1)})^2\,d\boldsymbol{x} + 2((\boldsymbol{q}^{(1)})^e)^T\boldsymbol{W}^e\boldsymbol{p}^{(1)}\\ &-|T|(\boldsymbol{q}^{(1)})^T\boldsymbol{W}((\boldsymbol{D}_\xi\boldsymbol{u}+\boldsymbol{D}_\eta\boldsymbol{v})\odot\boldsymbol{q}^{(1)}) \qquad (46)\\ \frac{d}{dt}(|T|(\boldsymbol{q}^{(2)})^T\boldsymbol{W}\boldsymbol{q}^{(2)}) = &-\oint_{\partial\mathcal{T}}(\boldsymbol{n}^{(2)}\cdot\boldsymbol{V})(q^{(2)})^2\,d\boldsymbol{x} + 2((\boldsymbol{q}^{(2)})^e)^T\boldsymbol{W}^e\boldsymbol{p}^{(2)}\\ &-|T|(\boldsymbol{q}^{(2)})^T\boldsymbol{W}((\boldsymbol{D}_\xi\boldsymbol{u}+\boldsymbol{D}_\eta\boldsymbol{v})\odot\boldsymbol{q}^{(2)}) \qquad (47)\end{aligned}$$

To enforce interface condition, we introduce the following flux and numerical flux functions:

$$\boldsymbol{f}(\boldsymbol{q}^-) = \boldsymbol{V}\boldsymbol{q}^-, \quad \boldsymbol{f}^*(\boldsymbol{q}^-,\boldsymbol{q}^+) = \frac{\boldsymbol{V}q^- + \boldsymbol{V}q^+}{2} + \frac{C}{2}(\boldsymbol{n}^-q^- + \boldsymbol{n}^+q^+) \qquad (48)$$

Note that if $C = 0$ then $\boldsymbol{f}^*$ is the central flux function. If $\boldsymbol{n}^- = \boldsymbol{n} = -\boldsymbol{n}^+$ and $C = |\boldsymbol{n}\cdot\boldsymbol{V}|$ then we recover the upwind flux function.

To enforce interface condition from the domain (1) side, we have, for $q^- = q^{(1)}$, $q^+ = q^{(2)}$, $\boldsymbol{n}^- = \boldsymbol{n}^{(1)}$, $\boldsymbol{n}^+ = \boldsymbol{n}^{(2)} = -\boldsymbol{n}^{(1)}$,

$$\boldsymbol{f}(q^{(1)}) = \boldsymbol{V}q^{(1)}, \quad \boldsymbol{f}^*(q^{(1)},q^{(2)}) = \frac{\boldsymbol{V}q^{(1)} + \boldsymbol{V}q^{(2)}}{2} + \frac{C}{2}\boldsymbol{n}^{(1)}(q^{(1)} - q^{(2)})$$

and to enforce interface condition from the domain (2) side, we have, for $q^- = q^{(2)}$, $q^+ = q^{(1)}$, $\boldsymbol{n}^- = \boldsymbol{n}^{(2)}$, $\boldsymbol{n}^+ = \boldsymbol{n}^{(1)} = -\boldsymbol{n}^{(2)}$,

$$\boldsymbol{f}(q^{(2)}) = \boldsymbol{V}q^{(2)}, \quad \boldsymbol{f}^*(q^{(2)},q^{(1)}) = \frac{\boldsymbol{V}q^{(2)} + \boldsymbol{V}q^{(1)}}{2} + \frac{C}{2}\boldsymbol{n}^{(2)}(q^{(2)} - q^{(1)})$$

For domain (1), we have

$$\begin{aligned}2((\boldsymbol{q}^{(1)})^e)^T\boldsymbol{W}^e\boldsymbol{p}^{(1)} &= \oint_{\partial\mathcal{T}} 2q^{(1)}\left(\boldsymbol{n}^{(1)}\cdot(\boldsymbol{f}(q^{(1)}) - \boldsymbol{f}^*(q^{(1)},q^{(2)}))\right)d\boldsymbol{\xi}\\ &= \oint_{\partial\mathcal{T}} 2\left(\boldsymbol{n}^{(1)}\cdot\boldsymbol{V}(q^{(1)})^2 - \boldsymbol{n}^{(1)}\cdot\boldsymbol{V}\frac{(q^{(1)})^2 + q^{(1)}q^{(2)}}{2} - \frac{C}{2}((q^{(1)})^2 - q^{(1)}q^{(2)})\right)d\boldsymbol{\xi}\\ &= \oint_{\partial\mathcal{T}}\left(\boldsymbol{n}^{(1)}\cdot\boldsymbol{V}\right)(q^{(1)})^2\,d\boldsymbol{\xi} - \oint_{\partial\mathcal{T}}\left(\boldsymbol{n}^{(1)}\cdot\boldsymbol{V}\right)q^{(1)}q^{(2)}\,d\boldsymbol{\xi}\\ &\quad - \oint_{\partial\mathcal{T}} C((q^{(1)})^2 - q^{(1)}q^{(2)})\,d\boldsymbol{\xi} \qquad (49)\end{aligned}$$

For domain (2), we have

$$\begin{aligned}2((\boldsymbol{q}^{(2)})^e)^T\boldsymbol{W}^e\boldsymbol{p}^{(2)} &= \oint_{\partial\mathcal{T}} 2q^{(2)}\left(\boldsymbol{n}^{(2)}\cdot(\boldsymbol{f}(q^{(2)}) - \boldsymbol{f}^*(q^{(2)},q^{(1)}))\right)d\boldsymbol{\xi}\\ &= \oint_{\partial\mathcal{T}} 2\left(\boldsymbol{n}^{(2)}\cdot\boldsymbol{V}(q^{(2)})^2 - \boldsymbol{n}^{(2)}\cdot\boldsymbol{V}\frac{(q^{(2)})^2 + q^{(2)}q^{(1)}}{2} - \frac{C}{2}((q^{(2)})^2 - q^{(2)}q^{(1)})\right)d\boldsymbol{\xi}\\ &= \oint_{\partial\mathcal{T}}\left(\boldsymbol{n}^{(2)}\cdot\boldsymbol{V}\right)(q^{(2)})^2\,d\boldsymbol{\xi} - \oint_{\partial\mathcal{T}}\left(\boldsymbol{n}^{(2)}\cdot\boldsymbol{V}\right)q^{(2)}q^{(1)}\,d\boldsymbol{\xi}\\ &\quad - \oint_{\partial\mathcal{T}} C((q^{(2)})^2 - q^{(2)}q^{(1)})\,d\boldsymbol{\xi} \qquad (50)\end{aligned}$$

The total energy rate equation s

$$\frac{d}{dt}\sum_{k=1}^{2}(|T|(\boldsymbol{q}^{(k)})^T\boldsymbol{W}\boldsymbol{q}^{(k)}) = -\sum_{k=1}^{2}\oint_{\partial\mathcal{T}}(\boldsymbol{n}^{(k)}\cdot\boldsymbol{V})(q^{(k)})^2\,d\boldsymbol{x} + \sum_{k=1}^{2}2((\boldsymbol{q}^{(k)})^e)^T\boldsymbol{W}^e\boldsymbol{p}^{(k)}$$
$$-\sum_{k=1}^{2}|T|(\boldsymbol{q}^{(k)})^T\boldsymbol{W}((\boldsymbol{D}_\xi\boldsymbol{u}+\boldsymbol{D}_\eta\boldsymbol{v})\odot\boldsymbol{q}^{(k)}) \tag{51}$$

Thus, at an interface point the energy is

$$-C((q^{(1)})^2 - q^{(1)}q^{(2)}) - C((q^{(2)})^2 - q^{(2)}q^{(1)}) = -C(q^{(1)}) - q^{(2)})^2$$

Thus, to ensure dissipation at the interface, we may choose C as follows,

$$C = \begin{cases} |\boldsymbol{n}\cdot\boldsymbol{V}| & \text{upwind flux} \\ \alpha\max_T|\boldsymbol{V}|, \quad \alpha\geq 1 & \text{Lax-Friedrichs flux} \end{cases} \tag{52}$$

We shown that if $C > 0$ the scheme is stable in the semi-discrete level. However, at the fully-discrete level, the scheme may not be stable, due to the amount of dissipation is not enough. Hence, we may adjust the value of α so that the scheme can be made stable. However, the larger the value of α, the smaller time step is required.

Octahedron Sphere